A discrete de Rham Scheme for an Interface Elliptic Problem: application to the Leaky Dielectric Model

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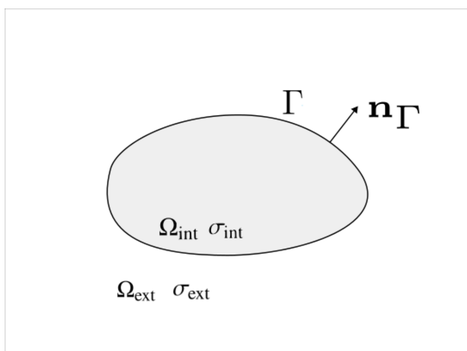
Model Problem

The domain $\Omega \subset \mathbb{R}^d$ is partitioned into $\Omega_{\rm int}$ and $\Omega_{\rm ext}$. Let $\sigma : \Omega \setminus \Gamma \to \mathbb{R}$ such that $\sigma_{|\Omega_{\rm int}} \equiv \sigma_{\rm int} > 0$ and $\sigma_{|\Omega_{\rm ext}} \equiv \sigma_{\rm ext} > 0$ be a diffusion coefficient discontinuous across Γ. Given a final time $t_{\rm f} > 0$, a source term $f = (f_{\rm int}, f_{\rm ext})$, with $f_\bullet : \Omega_\bullet \times (0, t_{\rm f}] \to \mathbb{R}$, $\bullet \in \{\rm int, ext\}$,the interface flux $\Phi_\Gamma : \Gamma \to \mathbb{R}$, $C \in \mathbb{R}^+$ and an initial potential jump J_Γ^0, we consider the problem of finding the time-dependent potential $u = (u_{\rm int}, u_{\rm ext})$ with $u_\bullet : \Omega_\bullet \times (0, t_{\rm f}] \to \mathbb{R}$, $\bullet \in \{\rm int, ext\}$ and the interface jump $J_\Gamma : \Gamma \times [0, t_{\rm f}] \to \mathbb{R}$ such that:

$$
\begin{aligned}
-\nabla \cdot (\sigma_\bullet \nabla u_\bullet) &= f && \text{in } \Omega_\bullet \times (0, t_{\rm f}],\ \bullet \in \{\rm int, ext\}, && (1a)\\
[u]_\Gamma &= J_\Gamma && \text{on } \Gamma \times (0, t_{\rm f}], && (1b)\\
[\sigma \nabla u]_\Gamma \cdot n_\Gamma &= \Phi_\Gamma && \text{on } \Gamma \times (0, t_{\rm f}], && (1c)\\
u_{\rm ext} &= 0 && \text{on } \partial\Omega \times (0, t_{\rm f}], && (1d)\\
C \partial_t J_\Gamma &= \sigma_\bullet \nabla u_\bullet \cdot n_\Gamma && \text{on } \Gamma \times (0, t_{\rm f}],\ \bullet \in \{\rm int, ext\}, && (1e)\\
J_\Gamma(\cdot, 0) &= J_\Gamma^0 && \text{on } \Gamma. && (1f)
\end{aligned}
$$



Physical Interpretation:
Two media with electric conductivity respectively equal to $\sigma_{\rm int}$ and $\sigma_{\rm ext}$ occupy the regions $\Omega_{\rm int}$ and $\Omega_{\rm ext}$. The interface Γ between the two media is characterized by a capacitance C. The variation rate of the charge in the bulk is f and the interface supports a surface charge Φ_Γ. The region $\Omega_\bullet$ with $\bullet \in \{\rm int, ext\}$ is characterized by an electrostatic potential $u_\bullet$ to determine. The potential is discontinuous at the interface, with a jump J_Γ to determine, and vanishes on the boundary of Ω.

Discrete Setting

Consider a polygonal mesh $(\mathcal{T}_h, \mathcal{E}_h)$, with $\mathcal{T}_h$ collecting the mesh elements and $\mathcal{E}_h$ the mesh edges. We additionally denote by $\mathcal{V}_h$ the set of vertices collecting the edge endpoints. The mesh is assumed to be fitted to the interface, i.e., there exists a subset $\mathcal{E}_{\Gamma,h}$ of $\mathcal{E}_h$ such that $\Gamma = \bigcup_{E \in \mathcal{E}_{\Gamma,h}} \overline{E}$.

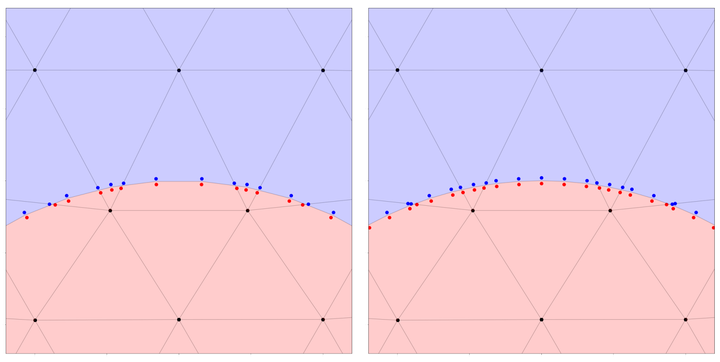


Figure 1: Admissible meshes with different refinement level of the interface.

- Fitted polytopal meshes can be obtained cutting the elements of a background mesh along the interface: no remeshing.
- Degradation of mesh quality can be countered through element agglomeration techniques leveraging the support of polytopal elements.
- The approximation of the interface is totally independent of the background mesh: the refinement level of the interface is arbitrary.

Space of Unknowns and Local Operators

For any $k \geq 0$ and $\bullet \in \{\rm int, ext\}$, we let

$$
\begin{aligned}
\underline{V}_{\bullet,h}^k := \Big\{ \underline{v}_h = \big((v_T)_{T \in \mathcal{T}_h^\bullet}, (v_E)_{E \in \mathcal{E}_h^\bullet}, (v_V)_{V \in \mathcal{V}_h^\bullet}\big) \ :\ & v_T \in \mathcal{P}^{k-1}(T) \text{ for all } T \in \mathcal{T}_h^\bullet,\\
& v_E \in \mathcal{P}^{k-1}(E) \text{ for all } E \in \mathcal{E}_h^\bullet,\\
& v_V \in \mathbb{R} \text{ for all } V \in \mathcal{V}_h^\bullet \Big\}.
\end{aligned}
$$

We consider the discrete space

$$\underline{V}_h^k := \underline{V}_{{\rm int},h}^k \times \underline{V}_{{\rm ext},h}^k.$$

We define the discrete gradient $G_T^k : \underline{V}_T^k \to \mathcal{P}^k(T)^2$ and potential $p_T^{k+1} : \underline{V}_T^k \to \mathcal{P}^{k+1}(T)$ such that, for all $\underline{v}_T \in \underline{V}_T^k$,

$$
\begin{aligned}
\int_T G_T^k \underline{v}_T \cdot \tau &= -\int_T v_T (\nabla \cdot \tau) + \sum_{E \in \mathcal{E}_T} \omega_{TE} \int_E v_{TE} (\tau \cdot n_E) && \forall \tau \in \mathcal{P}^k(T)^2,\\
\int_T p_T^{k+1} \underline{v}_T (\nabla \cdot \tau) &= -\int_T G_T^k \underline{v}_T \cdot \tau + \sum_{E \in \mathcal{E}_T} \omega_{TE} \int_E v_{TE} (\tau \cdot n_E) && \forall \tau \in \mathcal{R}^{{\rm c},k+2}(T). \qquad (2)
\end{aligned}
$$

We equip $\underline{V}_h^k$ with the energy norm:

$$\|\underline{v}_h\|_{{\rm en},h}^2 := \sum_{T \in \mathcal{T}_h} \sigma_T \left(\|G_T^k \underline{v}_T\|_{L^2(T)^2}^2 + h_T^{-1} \sum_{E \in \mathcal{E}_T} \|p_T^{k+1} \underline{v}_T - v_{TE}\|_{L^2(E)}^2 \right) + |\underline{v}_h|_{{\rm J},h}^2, \qquad (3)$$

where the interface jump seminorm is such that

$$|\underline{v}_h|_{{\rm J},h}^2 := \sum_{E \in \mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \|[\underline{v}_h]_E\|_{L^2(E)}^2. \qquad (4)$$

We also consider the space of interface jumps:

$$\underline{Z}_{\Gamma,h} := \Big\{ \underline{p}_{\Gamma,h} = (p_V)_{V \in \mathcal{V}_{\Gamma,h}} \ :\ p_V \in \mathbb{R} \quad \forall V \in \mathcal{V}_{\Gamma,h} \Big\}.$$

equipped with the norm:

$$\|p_{\Gamma,h}\|_{0,\Gamma,h}^2 := \sum_{E \in \mathcal{E}_{\Gamma,h}} \|p_E\|_E^2.$$

Stationary Jump Problem

If the interface jump is stationary, the weak formulation of the problem reads:
Find $u \in H^1(\Omega_{\rm int}) \times H^1_{0,\partial\Omega}(\Omega_{\rm ext}) = V_0$ such that

$$
\begin{aligned}
\sum_{\bullet \in \{\rm int, ext\}} \int_{\Omega_\bullet} \sigma \nabla u \cdot \nabla v - \int_\Gamma \{\nabla u \cdot n\}_k v + \eta \int_\Gamma [u][v] &=\\
\sum_{\bullet \in \{\rm int, ext\}} \int_{\Omega_\bullet} f v + \int_\Gamma \Phi_\Gamma \{v\}_{\overline{k}} + \eta \int_\Gamma J_\Gamma [v] \quad \forall v \in V_0, &
\end{aligned} \qquad (5)
$$

where η is a penalty parameter to impose weakly the jump condition and the average and skew-average operators are defined as:

$$\{v\}_k := \lambda_{\rm int} v_{\rm int} + \lambda_{\rm ext} v_{\rm ext}, \quad \{v\}_{\overline{k}} := \lambda_{\rm ext} v_{\rm int} + \lambda_{\rm int} v_{\rm ext},$$

with $\lambda_{\rm int}$ and $\lambda_{\rm ext}$ such that

$$\lambda_{\rm int} := \frac{\sigma_{\rm ext}}{\sigma_{\rm int} + \sigma_{\rm ext}}, \qquad \lambda_{\rm ext} := \frac{\sigma_{\rm int}}{\sigma_{\rm int} + \sigma_{\rm ext}}.$$

Discrete Version:

Let

$$\sigma_T := \sigma_{|T} \qquad \forall T \in \mathcal{T}_h.$$

We define the bilinear form $a_h : \underline{V}_h^k \times \underline{V}_h^k \to \mathbb{R}$ and the linear form $\ell_h : \underline{V}_h^k \to \mathbb{R}$ such that, for all $(\underline{w}_h, \underline{v}_h) \in \underline{V}_h^k \times \underline{V}_h^k$,

$$
\begin{aligned}
a_h(\underline{w}_h, \underline{v}_h) :=& \\
& \sum_{T \in \mathcal{T}_h} \left(\int_T \sigma_T G_T^k \underline{w}_T \cdot G_T^k \underline{v}_T + \frac{\sigma_T}{h_T} \sum_{E \in \mathcal{E}_T} \int_E (p_T^{k+1} \underline{w}_T - w_{TE})(p_T^{k+1} \underline{v}_T - v_{TE}) \right)\\
& - \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_h^k \underline{w}_h\}_{k,E} \cdot n_E [\underline{v}_h]_E + \eta \sum_{E \in \mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \int_E [\underline{w}_h]_E [\underline{v}_h]_E,
\end{aligned} \qquad (6)
$$

where

$$\alpha := \frac{2 \sigma_{\rm int} \sigma_{\rm ext}}{\sigma_{\rm int} + \sigma_{\rm ext}} \qquad (7)$$

and

$$\ell_h(\underline{v}_h) := \sum_{T \in \mathcal{T}_h} \int_T f p_T^{k+1} \underline{v}_T + \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \Phi_\Gamma \{\underline{v}_h\}_{\overline{k},E} + \eta \sum_{E \in \mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \int_E J_\Gamma [\underline{v}_h]_E. \qquad (8)$$

The discrete problem reads: Find $\underline{u}_h \in \underline{V}_{h,0}^k$ such that

$$a_h(\underline{u}_h, \underline{v}_h) = \ell_h(\underline{v}_h) \qquad \forall \underline{v}_h \in \underline{V}_{h,0}^k. \qquad (9)$$

- The jump condition is enforced *à la Nitsche*, leveraging on a user-dependent parameter η, taken large enough for stability.
- The scheme above is not a simple extension of discontinuous Galerkin techniques to the interface problem with the element potential playing the role of the discontinuous solution inside trace operators. On the contrary, such operators use the edge potential.
- The average is suitably weighted such that the properties of the method are independent of the conductivity contrast.

Main Results

Assume

$$\eta > \frac{C_{\rm tr}^2 N_\partial}{4\epsilon} \qquad (10)$$

for some real number $0 < \epsilon < 1$, with $C_{\rm tr}$ and N_∂ parameters depending only on the mesh quality. Then, for all $\underline{v}_h \in \underline{V}_{h,0}^k$, it holds

$$C_{\rm stab} \|\underline{v}_h\|_{{\rm en},h}^2 \leq a_h(\underline{v}_h, \underline{v}_h), \qquad (11)$$

with $C_{\rm stab} := \min\left\{1 - \epsilon, \eta - \frac{C_{\rm tr}^2 N_\partial}{4\epsilon}\right\}$.

Let u be the weak solution to (5) and $\underline{u}_h$ solve (9). Under assumption (10), and further assuming that $u \in \left[C^0(\overline{\Omega}_{\rm int}) \cap H^{r+2}(\mathcal{T}_h^{\rm int})\right] \times \left[C^0(\overline{\Omega}_{\rm ext}) \cap H^{r+2}(\mathcal{T}_h^{\rm ext})\right]$ for some $r \in \{0, \dots, k\}$, it holds

$$\|\underline{u}_h - \underline{I}_h^k u\|_{{\rm en},h} \lesssim \overline{\sigma}^{1/2} h^{r+1} |u|_{H^{r+2}(\mathcal{T}_h)},$$

where $|\cdot|_{H^{r+2}(\mathcal{T}_h)}$ is the broken H^{r+2}-seminorm on the mesh $\mathcal{T}_h$ and the hidden constant depends only on the domain, the stability constant $C_{\rm stab}$ in (11), the polynomial degree k, and the mesh regularity parameter (but is independent of both the meshsize and σ).

The right-hand side of the above estimate does not depend on the ratio $\frac{\sigma_{\rm int}}{\sigma_{\rm ext}}$, making it robust in the case of media with highly contrasting properties. Crucial to obtain this robustness property is the use of weighted averages.

Interface Jump Time Discretization

Introduce a time step τ and define $J_\Gamma^n = J_\Gamma(n\tau)$; J_Γ^{n+1} is determined by the time-discrete variational version of Equation 1e: Find $J_\Gamma \in L^2(\Gamma) = Z$ such that

$$\frac{C}{\tau} \left(\int_\Gamma J_\Gamma^{n+1} Q_\Gamma^n - \int_\Gamma J_\Gamma^n Q_\Gamma^n \right) = \int_\Gamma (\sigma_{\rm int} \nabla u_{\rm int}^n \cdot n_\Gamma) Q_\Gamma^n = \int_\Gamma (\sigma_{\rm ext} \nabla u_{\rm ext}^n \cdot n_\Gamma) Q_\Gamma^n \quad \forall Q_\Gamma \in Z.$$

Discrete Version:

Find $\underline{J}_{\Gamma,h}^{n+1} \in \underline{Z}_{\Gamma,h}$ such that:

$$\frac{C}{\tau} \left(\sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E J_E^{n+1} Q_E^n - \int_E J_E^n Q_E^n \right) = \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_T \underline{u}_T\}_{k,E} \cdot n_\Gamma Q_E \qquad \forall \underline{Q}_{\Gamma,h}^n \in \underline{Z}_{\Gamma,h}. \qquad (12)$$

Time Discretization Algorithm

Given $\underline{J}_{\Gamma,h}^n \in \underline{Z}_{\Gamma,h}^k$, denote by $\mathcal{M}_h : \underline{Z}_{\Gamma,h}^k \to \underline{V}_h^0$ the operator that associates to a jump $\underline{J}_\Gamma^n$ the solution $\underline{u}_h$ of the stationary problem (9). Likewise, call $\mathcal{N}_h : \underline{V}_h^0 \times \underline{Z}_{\Gamma,h} \to \underline{Z}_{\Gamma,h}$ the operator that associates to a potential $\underline{u}_h$ and a jump $\underline{J}_{\Gamma,h}^n$ the jump $\underline{J}_{\Gamma,h}^{n+1}$ solution to problem (12). Then, the time advancement algorithm for the case of an evolving jump reads: Given $\underline{J}_{\Gamma,h}^0$, for $n = 0, \dots, N-1$, set, in this order,

$$
\begin{aligned}
\underline{u}_h^n &= \mathcal{M}_h(\underline{J}_\Gamma^n),\\
\underline{J}_{\Gamma,n}^{n+1} &= \mathcal{N}_h(\underline{u}_h, \underline{J}_{\Gamma,h}^n).
\end{aligned}
$$

Numerical test: stationary jump problems

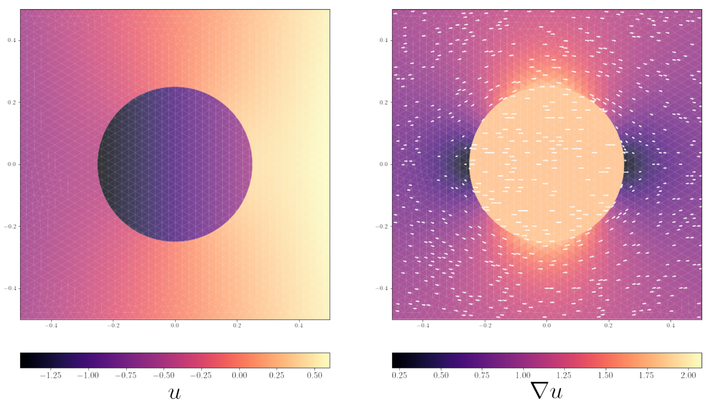


Figure 2: Numerical solution and gradient with circular interface.

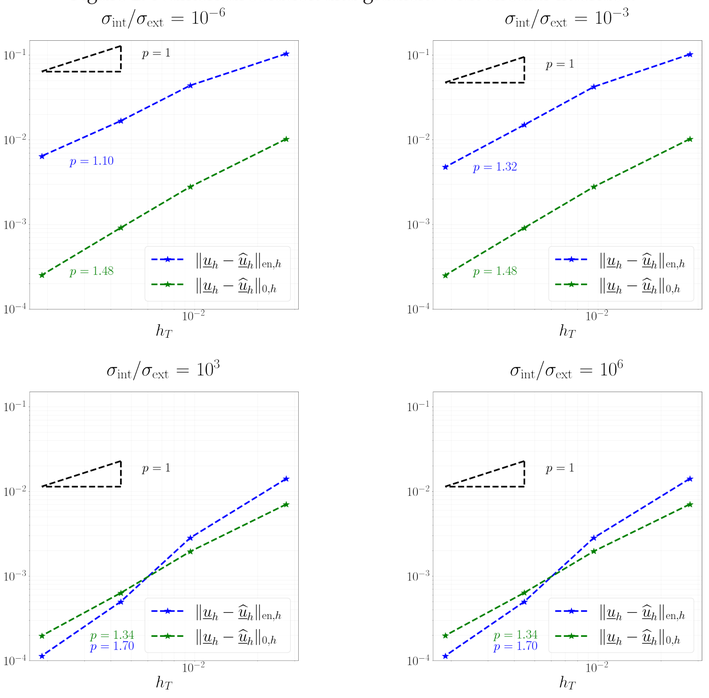


Figure 3: Check on the indepedence of the convergence ratio from the conductivity ratio.

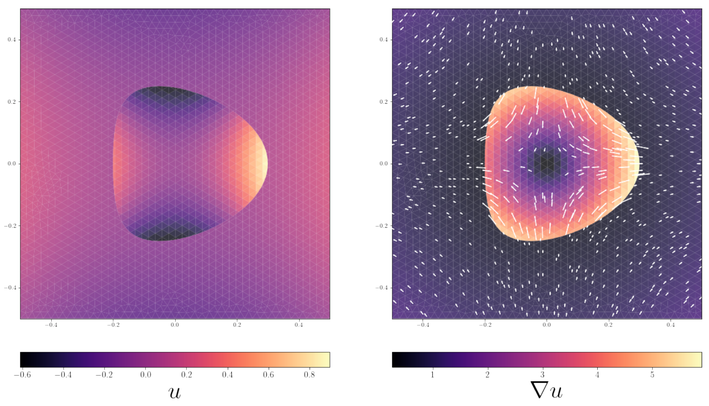


Figure 4: Numerical solution and gradient with generic interface.

Numerical test: time-dependent jump

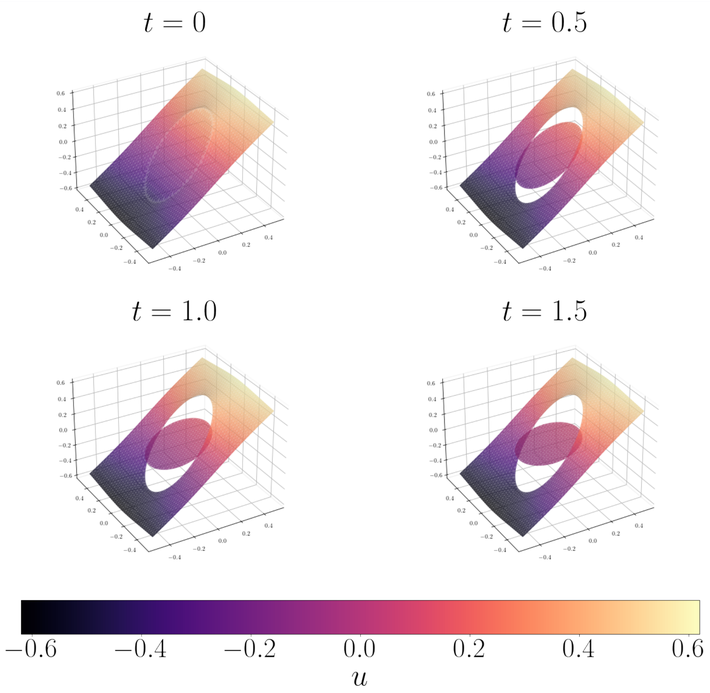


Figure 5: Loading process of a circular capacitor subject to a uniform electric field.

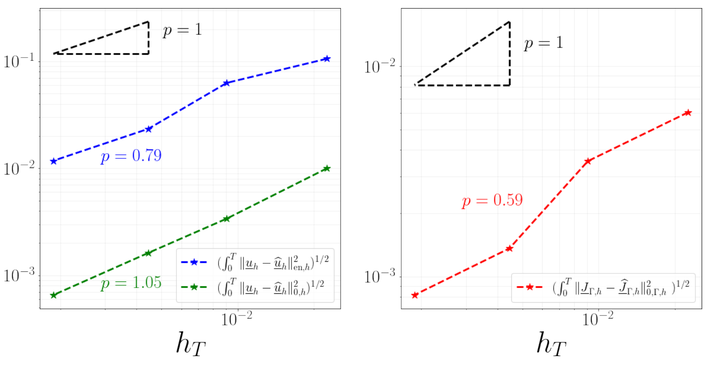


Figure 6: Mesh convergence study for the time-dependent problem.

Bibliography & Acknowledgement

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